

Resume: Ankit Jain
(Senior Manager in JCB).

(B.Tech. NIT Calicut).
EX Employee: (TATA/Mahindra/John Deere)

+ 14.5 years in New Product development & Innovation, Business Strategy, Program Mgmt., VAVE Eng., Empathetic Leader, Communicator, Execution, legislation, Testing & Automobile

Contact Details: +91- 9356152473/8669962188; Email/Linked In: ankitjain071@gmail.com

Technical SKILLS:

Power Train, Engine & Engine Aux, Structure, Hydraulics, Innovative Vehicle design and development, Vehicle Integration, Product Management, Program Management, Durability & Validation, Product Portfolio, Product Positioning with Respect to Market and customer Requirement, Automobile and off highway Rich Experience etc.

Managerial Skills:

Out of the box thinking, Managing Team/counter parts/CFT, Challenging Environment and goals, Developing skills in team, multi-tasking/projects with Resource utilization, Capex, Budget, Conflict Management, Rewards and Recognition, People Management etc.

Expertise Skills:

Domestic/International Product handled, UK-USA cross functional team Exposure, MNC – John Deere, TATA, Mahindra, JCB, Automobile- off highway Products hands on Backhoe Loader, Wheel Loading Shovel, Motor Grader, Pick and Carry Crane, Excavator. Departmental Experience- Program Management, Design, Product development, Engineering Manager

Vehicle Design: Vehicle layout/New technology shifts/Product Positioning in Market/Mc Specification/Engine-Power Train Design-Selection/Hydraulics Schematics & Parameters/DFMEA/Project Attributes/Performance Deliverables/Standard Mc configuration/Base Mc BOM costing/Optional-Premium Features, Clean Packaging & SAE Targets, Total cost of Ownership.

DVPSOR: (Internal/External) performance/endurance required at Component/Vehicle level. Customer Evaluation/attributes measurement, Measuring usage pattern (Duty Cycles) on product life cycle, Data Mapping, Application requirement.

Engineering Process: Machine BOM configuration ECN/DFMEA/DFM/SOR/DVPSOR/BOM/SOP/Design guidelines/Service requirements/Sequence of Assembly, Supplier SOP etc.

Product Benchmarking: Feature/Service/Specification/Competitive Product/VAVE/Failure Modes/Attributes benchmarking, Requirement for Specific application, Improvement in existing products, New Technology, Productivity/profitability/Sourcing.

Delivering outcomes: Out of the Box Thinking, problem solving skills, Execution skills, proactive approach, 8D-Why-Why analysis, prioritizing projects, Daily Tracking, engaging team, resource utilization, data management, decision-making ability

Supplier Development: New design/Technology development, SOR-Design Specification-Validation/Supplier selection

Manager (Leadership Skills): Define outcome, team direction-goals (KRA), Project allocation, Team engaging, Daily tracking, Team skills, Team utilization, Multiple projects/tasks enhancing, Effective communication & positive Environment approach.

Product Attributes: Risk Mitigation/Parallel timelines/ Quality targets/small-large Scale projects/CFT Responsibilities/Counterparts UK, America, China alignment Reviews.

EXECUTIVE SUMMARY: PROFESSIONAL EXPERIENCE

(JCB INDIA PVT LTD): Senior Engineering Manager (2024 to Present); Manager (2018-2023) on Wheel Loading Shovel Platform (New product development, Care and Maintenance)

- ✓ Managing design Team, Design scope, New NPIP projects, T1/T3 quality targets, VAVE projects, setting KRA goals, Business continuity
- ✓ Manager on WLS NPIP (3T to 5.5 T) wheel loading shovels, BS5 emission regulation projects, current projects C&M, Export Projects for T3/T4 Markets Project, New Tenders Requirement, cost targets, design durability and product life cycle
- ✓ Engineering-Marketing Brief for product specification/attributes market positioning, continuity of JCB as No. 1 in wheel loading shovel business, legislation, domestic CMVR requirements, tackling Assy line Issues, Field customer issues, Supplier technology changes, product technology shift etc.
- ✓ Delivered 433/437/455 (3.3T to 5.5 T) BS4 Emission Products/AIS 160 phase I with Improved Fuel Efficiency targets & higher payloads and new platform for Wheel loading shovel with commonality design across product portfolio, Increasing penetration in Market for 3.3T WLS by 500 %.
- ✓ Delivered Defence tenders, field DEF related quality issues, DEF inline filtration technology, Structure related failures, Service-related concerns on Current BS4 Products
- ✓ Delivered 433/437/455 (3.3T to 5.5 T) BS5 Emission Products/AIS 160 phase II with Improved Fuel Efficiency targets & Cost targets with VAVE and design optimization with minimal changes in Existing Product, Improved margin by changing product 3.7T to 4T with desirable attributes, 424 North America premium features inside cabin, -20 deg ambient requirement cold start, Styling Features and converting common product for different region of world to increase Market volume/localization of components, reducing cost of ownership
- ✓ Vehicle Integration check, Design guideline, Team Management, Developing team, Design support on Assy Line/Manufacturing/Supplier development, Installation Requirement of New AFT installation/New Engine/New hydraulics/New cooling/structures... etc. along with AIS160 new Government Requirements
- ✓ line manager/Program-product manager, achieving timelines, different program gates, product lifecycle, defining DVPSOR, seeding Machine in fields, Operator/customer clinic for new product, competitive benchmarking, Machine USP, machine BOM, Marketing alignment, DFMEA, SOR etc.
- ✓ Meeting and Selection of Critical Driveline components (ZF, DANA, Lonking, Fleetgaurd, Donaldson, JCB engine, Tyre manufacturer, Rims, Vehicle Specification, Cycle time, Engine Power Eco Curve, Hydraulics parameter, break out and stability of Product during different application, Cabin Std-Option premium Features, Structural modification engaging FEA and Strain gauging, E&E Integration checks
- ✓ Defining Validation Plan, Endurance hours, fatigue and dynamic loading, Vehicle FAW/RAW, Component No. of cycles for B10 life, Supplier DVPSOR, Vehicle Testing inputs for Performance parameter, fine tuning of Engine torque requirement, Root cause analysis for failures and durability issues and mitigate program with No Risk
- ✓ CFT (Purchase/ME/Quality/Mfg.) inputs on Design change management, CTQ parameters, Assy requirement, BOM clarity-Vehicle configuration, End of Line Testing by defining SOP for smooth introduction of new product on each Program gates.
- ✓ Meetings: Daily War room with CFT, Design Solution/Decision taking ability/Team responsibility and delivery
- ✓ Design Counter Parts: Aligning JCB UK, US and China team, Design Review Technical Review internal and External, Data management, Documentation, Design guideline-calculations, Executive presentation, cost targets review, Capex, and forecasting of budget.
- ✓ Expertise in Designing parameters: *Engine Selection*, Power- Eco Mode torque selection, Aftertreatment Selection/Integration, DEF/AdBlue along with Cold Start/Hot, Engine Peripheral : Air intake system Design, Cooling system (Radiator, hydraulic, transmission, Fuel Cooler along with & Optimized Fan Selection, *Transmission selection (Manual/Torque-converter/Transaxle/HST) selection*, *Exhaust system installation (EGR, NVH, DOC, and SCR) selection*, *Fuel System (CRS, Mechanical engine, schematic layout)*, *Brake system (pneumatic, hydraulic, parking)*, *Clutch (diaphragm type) and Tire-Rim, Engine Mounts, propeller shaft, levers and controls. Hydraulic component selection, Breakout/Speed/Tractive effort/MRV/ARV & tipping load calculation to define vehicle specification*
- ✓ Quality matrix of new products/Product range/ Statistical/quantitative analysis to take decision on design solutions for failure analysis, Life estimation, vehicle duty cycles and durability.

(Mahindra & Mahindra): Engineering Manager (Aug-2017-October-2018); Deputy Manager (Feb-2014 to Aug2017) on Powertrain Lead, drivelines, Engine-Engine Aux on Backhoe Loaders, Motor Grader, Cranes etc.

- ✓ Delivered New Product Motor grader G75/G90 (power rating), 91 HP Backhoe Loader, 12T 7& 14 T hydra cranes, Concept of Haulage tractor by converting Backhoe, innovations/Lead in power train system.
- ✓ BS4 Emission regulations/Packaging on Existing 76/91 HP Backhoe loader/Motor Grader with 76/91 HP rating considering with/without SCR, Packaging of AFT components (DOC, DPF, SCR) with Adblue System, costing of Project, product attributes, styling, & validation plan.
- ✓ Engine-Engine peripherals selection, driveline selection, vehicle performance, calculations on gradability, drawbar pull, Stall ratio and productivity requirement, Product establishment in market.
- ✓ Delivered 91 HP BHL along with HVAC integration on Engine, HVAC, Parasitic and Mass requirement in power train domain, continuously involved in cost reduction and Field failure resolution under Mahindra 12 MIS/RPH target.
- ✓ Delivered 95Hp CRS Engine technology implementation on backhoe loader with engine calibration team for digitization/Optimization of productivity in BS3 emissions to enhance product portfolio and Fuel consumption targets.
- ✓ Delivered low NVH targets on BHL/Grader in Exhaust, Sound absorber, fan RPM Power Management, Exhaust low and high frequency, Silencer Chambers to reduce the vehicle noise by 4 dBA.
- ✓ Lead cold start kit development in BHL for India market, Defence tender requirement.
- ✓ Reduced field failures, Cost VAVE on New products Mobile Cranes, Excavator, Motor Grader with B10 life & performance targets
- ✓ Capacity to mentor, motivate and develop team to attain higher levels of performance and in building the competency.
- ✓ innovative Franna Cranes, 50 HP small BHL with Tractor Engine in conceptualization phase.
- ✓ Capacity to mentor, motivate and develop a team to attain higher levels of performance and in building the competency.
- ✓ Knowledge in world class product development process including DFMEA, DVP, SOR, PFMEA & Failure analysis.

(John Deere): Design Engineer/Senior Design Engineer on Excavators (June 2011-Jan 2014)

- ✓ Delivered powertrain design for New Excavators 14T & 16 T with CRS engine for China market, Engine and Engine Peripherals (Cooling System, Air intake, Engine Selection, Power Mapping with Variable Pump, Design quality check, FEA analysis, CFD analysis, Validation Plan
- ✓ Delivered Counter Parts: US Design Review, China Design Review, Manufacturing and Assy Concerns, Design on all NPIP gates
- ✓ Lead cost reduction project for existing excavators 21 and 24 T for china market in structure, localization of parts, Resourcing etc
- ✓ Engine selection, driveline selection, vehicle performance and Engine aux design core competency
- ✓ Member of COE/SME on Air Intake system/Cooling System

(TATA Auto comp): Assistant Manager TATA AUTOCOMP for (NPD) (GET) (July 2010-MAY2011)

- ✓ Handling Various New project/change management under EPDP phases till development stage complete to meet Timeline along with all cross functional Team.
- ✓ Identifying and resolving major critical activities under project, Preparing Timeline in MS Project, and Excel to follow up all tasks under various scenarios.
- ✓ Injection moulding components: I have looked out components of two projects Super Ace and World Truck. Successfully delivered components on time to meet time line and cost, Special training on Program Management, and learn on Time line management, utilizing Resources, etc. to get the things by push-pull strategy. Importance of CFT task /multiple tasks to crunch the timeline and PPAP documentation. Rotation for 2 months in purchase department for developing new parts.

Technical Expertise

1. Engine Selection: Power/Torque/SFC curve on vehicle Performance (Tractive effort, drawbar pull, Vehicle speed at each gear, gradability at each gear, Fuel/oil consumption for different applications, hydraulic power requirements and Engine Aux drive Requirement for off road vehicles. Parasitic calculation, Productivity and Fuel Efficiency operating range, Rpm stall ratio adjustment.
2. Cooling System: Radiator/Oil Coolers/Charge air cooler: Heat rejection/LAT product requirements, Optimize Airflow Requirement, Selection of Fins type, pressure drop across cooler, Staggered and inline coolers, Recovery bottle Capacity Selection, Isolators for cooler mounting.

3. Fan selection: Optimization of airflow across cooler, pressure drop across fan, Mechanical sucker and puller fan, direct drive fan for low cost, viscous fan for low noise & power consumption, Viscronic and Hydraulic fan, Fan Noise targets, Integration of fan
4. Air Intake/pre-cleaner selection: based on Airflow and pressure drop requirement. Selection based on efficiency at coarse and fine dust application, Dust holding capacity requirements, indications for serviceability, single and dual filter elements. Fixed vane/turbo blade type, scavenging process for self-cleaning.
5. Exhaust System: DOC/DPF/SCR/AdBlue back pressure target, noise level targets, Silencer with multiple chambers, flute and perforation concepts, acoustic material, Integration, and heat shields.
6. Tyre/Rim Selection: Load carrying capacity requirements, slip ratios for 2WD/4WD modes, static and dynamic loading radius, Trade/PLY selection for minimum Tyre wear, Tyre selections based on product application, single/split rim design.
7. Transmission/Transaxle System: manual/power shift transmission, no. of gear shift/gear ratio selection, torque converter for more torque ratio, Vehicle level performance, integration of propeller shaft, trouble shooting, electronics solenoids, pump requirements, Gear shift lever design based on ergonomics and Efforts.
8. Fuel System/Tank: Fuel tank capacity based on application & BSFC curve, vehicle Load Factor, Tank robust design, fuel layout, routing, Low pressure circuit, Bosch Fuel layout guideline for CRS and mechanical engine/Fuel filtration requirement (Fuel Filter, water separator, fuel bowl heater, DFHC, Fuel sender for fuel level mapping, Refill Pump, (continuous & intermittent requirement)
9. Brake design & layout: Pneumatic brake circuit, hydraulic brake circuit, parking brake.
10. Brake Component selection: Compressor, un-loader valve, air reservoir capacity, brake pedal valve, air booster, hydraulic master cylinder, pneumatic brake hose, Bundy tubes, brake fluid reservoir capacity based on pressure mapping and braking torque requirement to meet ISO std. (brake drum/ brake shoe, disc brake)
11. Clutch/Torque converter: Selection of clutch and integration with flywheel/transmission, clamp load and wear rate, Clutch pedal design based on Effort and reliability. TC torque ratio selection based on tractive effort requirement
12. Strain gauging/FEA: FEA based on different load factors, static-dynamic g loading, Vibration analysis requirement along with endurance on define duty cycles
13. NVH: Selection of AVM for isolation efficiency, Frequency, hand body vibration/ Engine AVM & noise attenuation with acoustic material, Fan RPM, Exhaust System, Shafts, hydraulic jerks, smooth ride system etc...
14. BSIV/FT4/BSV emission norms. EGR, DOC, SCR, DEF, CRDI for meeting different PM and NOx requirements. Integration of AFTD along with Bosch/Donaldson/Faurecia guideline and Service requirements of components. Regeneration/hot summer/winter/altitude requirement

Automobile Tools

PRO E/CATIA/UG/ VIS MOCK UP/JT2Go	New and Complex design and Validation, Complex design under part/sheet metal;/surface design/Routing/Assembly for vehicle. Hands on CATIA, Team centre and PLM management, Pro-E and PLM management from last 10 years
--------------------------------------	---

Educational & Personal Details

Course	UNIVERSITY	YEAR OF PASSING	CGPA/ Percentage
B. TECH (Mechanical Eng.)	National Institute of Technology, Calicut	2006-2010	7.9 CGPA
Class XII	M.P. State Board	2004	79.2 %
Class X	M.P. State Board	2002	79.6 %

Personal Details

Date of Birth:	04/07/1986
Gender/Marital Status	Male/Married
Hobbies	Cooking/Indoor games

I hereby solemnly declare that all the information furnished above is true to the best of my knowledge